

Varad Rane

NYC Metro Area | hello@varadrane.com | LinkedIn | GitHub | Blog | +1 201 268 8948 | Authorized to work in the US

SUMMARY

Backend engineer, 3 years production experience across trading infrastructure and applied LLM systems. Built token-based auth, an API gateway and production monitoring for trading APIs serving 2M users at one of India's largest brokerages. Shipped a RAG and document-extraction pipeline now handling 97% of client intake at a healthtech startup, plus a serverless AWS extraction pipeline on Step Functions, Bedrock and Aurora/pgvector with full Terraform IaC.

EXPERIENCE

- Software Developer** — *FieldWorker (Contract Position)* Princeton, NJ · Feb 2026 - Present
 - Built an LLM document-extraction pipeline (GPT-4.1) turning multi-column, cross-page PDF reports into schema-validated JSON, **automating ~95% of client intake** and removing manual data entry from the workflow.
 - Architected a local-first RAG pipeline for clinical case notes (Ollama, ChromaDB, sentence-transformer embeddings, PostgreSQL vector storage) engineered to **run inside an 8GB VRAM budget** so sensitive data never left the environment; adopted for **97% of new client entries in production**.
 - Designed a reproducible benchmarking harness measuring latency, throughput and retrieval quality across chunking, retrieval and quantization strategies; heading-based markdown chunking won, **raising retrieval accuracy 72% to 93%**.
- Teaching Assistant** — *New Jersey Institute of Technology* Newark, NJ · Sep 2025 - Dec 2025
 - Built an automated grading system running student Python and C++ submissions against test suites in isolated containers, with an LLM generating structured HTML feedback on code quality; cut **turnaround for 50 students from 5 days to 6 hours**.
- Software Engineer II** — *Kotak Securities* Mumbai, India · Jun 2023 - Aug 2024
 - Built **token-based authentication from scratch** for trading APIs serving 2M users at **~260 req/s peak, 30% above** the previous 200 req/s, setting the security and audit-compliance baseline for all later internal applications, **remediated 12 penetration-test findings** and authored the compliance docs.
 - Architected alert management across **100+ production servers and message queues** with configurable health checks, cutting quarterly downtime from **6 hours to 5** and automating post-incident root-cause reporting.
 - Built a .NET Standard 2.1 shared library for data access and logging adopted by **50+ engineers across 4 teams**, removed two network hops per request and resolved a production IIS socket-exhaustion failure.
 - Designed an internal API gateway **centralizing auth, CORS and rate limiting, with SQL-driven routing** that enabled parallel live-environment testing against production data with no separate staging deployment.
- Software Engineer** — *Kotak Securities* Mumbai, India · Jun 2022 - Jun 2023
 - Optimized high-frequency production SQL via execution-plan analysis and indexing, **cutting p95 latency** on critical retrieval paths from **237 ms to 145 ms**.
 - Migrated a legacy CRM (70 views) from **.NET Framework to .NET 7** with zero production downtime, resolving breaking changes and realigning auth flows to updated security standards.
 - Built a **reusable bulk file-upload module** with validation, transactional error handling and user-facing recovery, **standardizing CSV ingestion** across 7 teams and halving development time for new upload workflows.

PROJECTS

- Cloud Document Extraction Pipeline** — Python, AWS Lambda, Step Functions, Bedrock, Aurora + pgvector, Terraform
 - S3, Step Functions (Distributed Map), 8 Lambdas, Bedrock (Claude 3.5 Sonnet vision + Titan Embeddings), Aurora + pgvector
- PrivGuard Sentinel** — Python, GitHub Actions, LLM APIs
 - LLM pull-request reviewer posting inline comments from structured JSON. Chunks diffs past model context limits.
- LazyCurl** — Go, Bubble Tea
 - Lazygit-style terminal UI for HTTP APIs: session management, visual request editing, and a concurrent load-testing engine.

EDUCATION

- New Jersey Institute of Technology** — *MS, Computer Science · Jan 2025 - May 2026 · GPA 3.90 / 4.00* Newark, NJ
- University of Mumbai** — *BE, Electronics Engineering · Jun 2018 - May 2022 · GPA 9.32/10.0* Mumbai, India

SKILLS

Languages	Python, Go, C#, SQL/T-SQL, TypeScript, JavaScript, C++
Backend	REST API design, API gateways, microservices, distributed system design, OAuth 2.0, JWT, 2FA, IAM, rate limiting, ASP.NET Core, .NET Core/Standard, Node.js
Data	PostgreSQL, SQL Server, pgvector, ChromaDB, query optimization, execution-plan analysis, indexing, schema migration
Cloud	AWS (Lambda, Step Functions, Bedrock, Aurora, S3, API Gateway, CloudWatch), Terraform, Docker, GitHub Actions, CI/CD, Linux, observability
AI / LLM	RAG, embeddings, semantic search, vector DBs, prompt orchestration, structured output, LLM evaluation & benchmarking, chunking, quantization, Ollama
Other	React, Next.js, Angular, system design, code review, unit & integration testing, Agile

CERTIFICATES AND PUBLICATIONS

- AWS Building Production-Ready AI Agents with Amazon Bedrock AgentCore (2026)
- Rane, V., et al. "Automated Crack Severity Level Detection and Classification for Surface Crack Using Deep Convolutional Neural Networks." Springer: Advanced Computing and Intelligent Technologies, Aug 2022